



UNIVERSITI TEKNIKAL MALAYSIA MELAKA

FACULTY OF ARTIFICIAL INTELLIGENCE AND CYBERSECURITY

BAXZ2413 NETWORK SECURITY INFRASTRUCTURE AND DESIGN

PROFESSOR MADYA TS. DR. MOHD FAIZAL BIN ABDOLLAH

2 BAXZ

S1G2

PROJECT

No.	NAME
1.	FAEZ BIN NAZARI
2.	AYISHAH AMIRA BINTI MOHD ZAMRI
3.	WAN NORAZMEERA BINTI WAN JAFERI
4.	NUR NAJIAH TASNIM BINTI MOHAMAD FAIROS
5.	AIN NURIN BINTI MOHD ZAHIDIE
6.	NUR AINA SYUKRINA BINTI RUSLAN
8.	LIVEKHA A/P K NAGASUNDRAM
7.	SITI NORBAIYAH BINTI AN SARY
9.	NUR BATRISYA AMIRA BINTI MOHD RIDZUAN

1. Introduction and objectives

1.1 Introduction

Organizations today face increasing cybersecurity threats that require effective monitoring, centralized authentication, and rapid incident detection. Traditional standalone systems often lack visibility into security events occurring across multiple platforms, making it difficult for administrators to identify and respond to attacks efficiently.

To address these challenges, this project implements a centralized security infrastructure consisting of a Windows Server running Active Directory Domain Services, an Ubuntu Linux client integrated with the domain, an Intrusion Detection System (IDS), and the Wazuh Security Information and Event Management (SIEM) platform.

Active Directory provides centralized identity and access management, allowing users to authenticate using a single set of credentials across different operating systems. Ubuntu is configured to authenticate against Active Directory using SSSD and also hosts an IDS to monitor network activities. Security events generated by the operating system, authentication services, and IDS are collected and forwarded to Wazuh for centralized analysis and visualization. The implementation demonstrates how enterprise environments can integrate authentication, intrusion detection, and log analysis to improve overall security visibility and incident response capabilities.

1.2 Objectives

The objectives of this project are:

- To deploy and configure Active Directory Domain Services on Windows Server.
- To create and manage domain user accounts for centralized authentication.
- To install and configure an intrusion Detection System (IDS) on Ubuntu.
- To deploy Wazuh SIEM for centralized log collection and analysis.
- To evaluate the effectiveness of IDS and Wazuh in detecting and analyzing security threats.

2. System Architecture Diagram

2.1 Architecture Overview

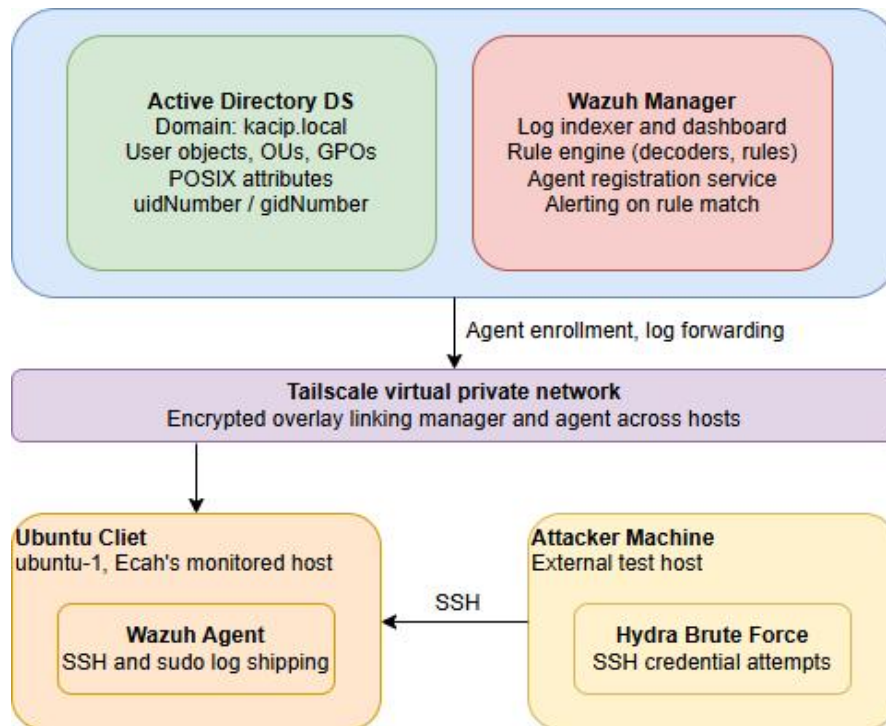


Figure 2.1 Architecture Overview

2.2 Architecture Description

The Windows Server hosts Active Directory Domain Services and DNS services, which provide centralized authentication and user management. Ubuntu Linux joins the Active Directory domain through SSSD, enabling domain users to log in using their AD credentials. An Intrusion Detection System (IDS) is deployed on Ubuntu to monitor network traffic and identify suspicious activities such as brute-force attacks, unauthorized access attempts, and abnormal traffic patterns. The IDS generates security alerts whenever malicious behavior is detected.

The Wazuh Agent installed on Ubuntu collects system logs, authentication logs, and IDS alerts before forwarding them to the Wazuh Manager. The Wazuh platform performs log analysis, event correlation, and alert generation while presenting security information through a centralized dashboard.

3. Methodology and configuration steps

3.1 Active Directory Configuration

The Windows Server was configured as a Domain Controller by installing the Active Directory Domain Services role. A new domain named "kacip.local" was created to manage users and authentication services. A domain user account named "john" was created through Active Directory Users and Computers (ADUC).

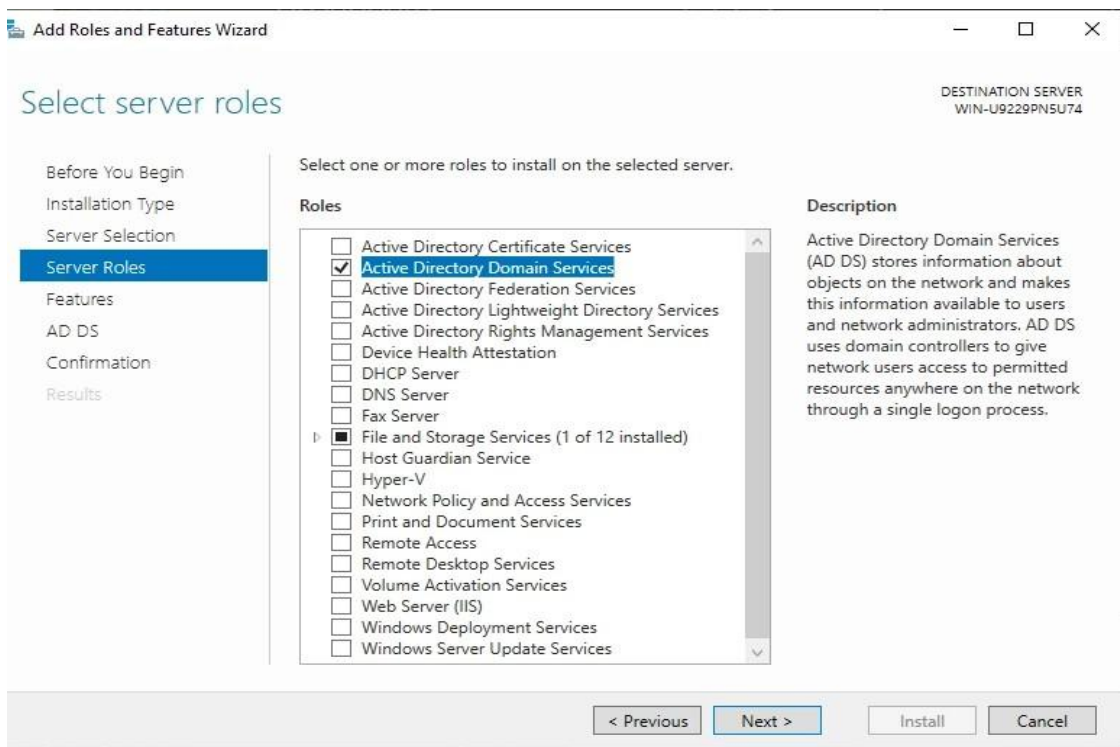


Figure 3.1.0: Selecting the Active Directory Domain Services (AD DS) role for installation

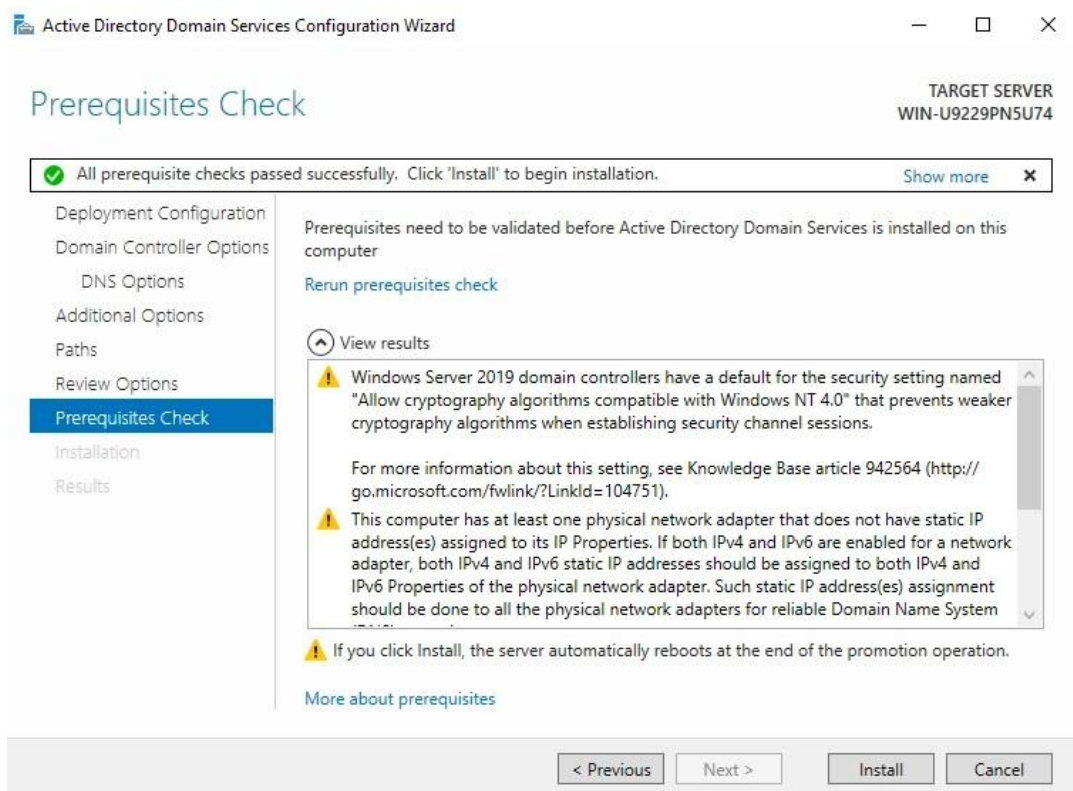


Figure 3.1.1: Successful prerequisite validation before Active Directory Domain Services (AD DS) installation.

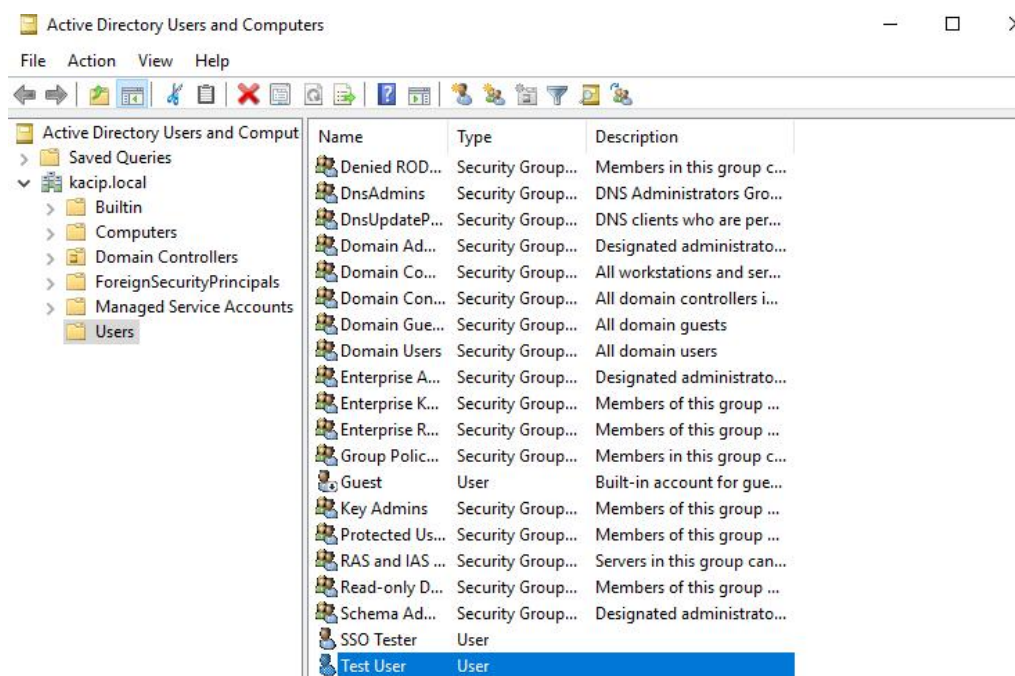


Figure 3.1.2: Successfully creation user account (Test User) in Active Directory Users and Computers (ADUC)

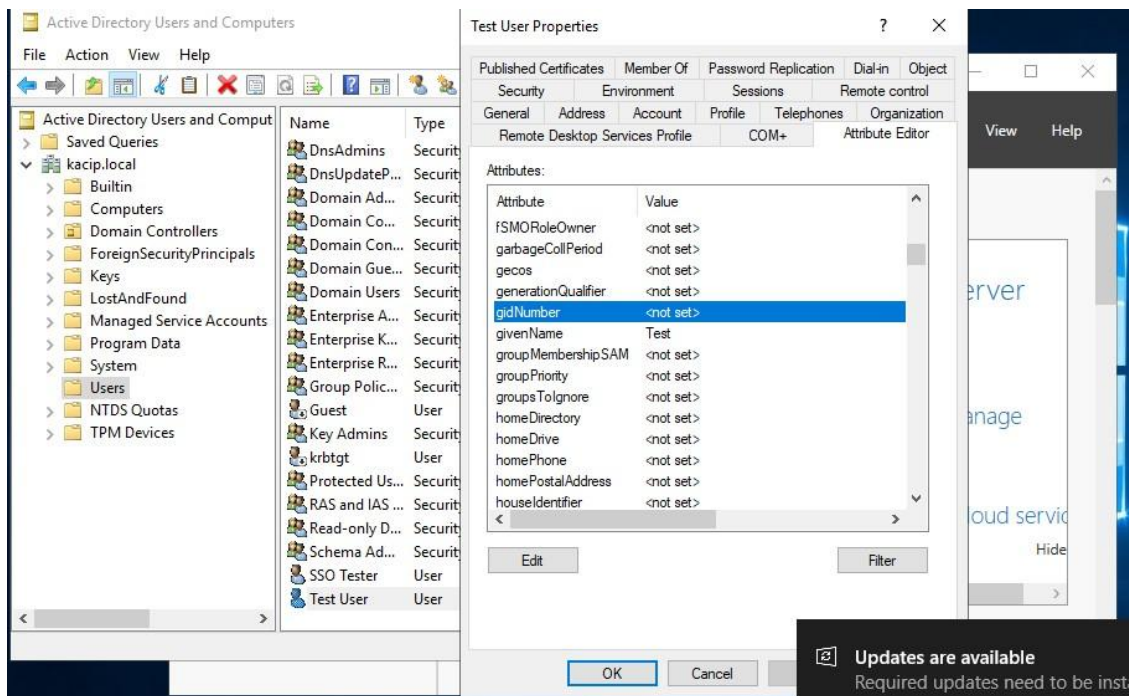


Figure 3.1.3: Viewing the properties and attributes of the Test User account in Active Directory Users and Computers (ADUC)

3.2 Ubuntu Domain Integration

Ubuntu was configured to communicate with Active Directory using the following components, which are Realmd, SSSD and ADCLI. The system was joined to the domain "kacip.local" and tested by authenticating with the domain user account "john". Successful authentication verified that Ubuntu could retrieve identity information directly from Active Directory.

```
ecah@ubuntu:~$ sudo apt update && sudo apt install realmd sssd sssd-tools libnss-sss libpam-sss adcli samba-common-bin o
ddjob oddjob-mkhomedir packagekit
[sudo] password for ecah:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,696 kB]
Get:5 https://pkgs.tailscale.com/stable/ubuntu noble InRelease
Get:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:7 https://pkgs.tailscale.com/stable/ubuntu noble/main amd64 Packages [14.8 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2,003 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [267 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [22.0 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2,997 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [697 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1,188 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [229 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [39.3 kB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [8,564 B]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [354 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [3,193 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [739 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,689 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [328 kB]
Get:24 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:25 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [44.7 kB]
Get:26 http://archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [10.7 kB]
Get:27 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
```

Figure 3.2.0: Ubuntu terminal where a user is preparing the system to join a Windows Active Directory (AD) domain by installing the required package dependencies.

```

4 RestrictedKrbHost/UBUNTU@KACIP.LOCAL
4 RestrictedKrbHost/UBUNTU@KACIP.LOCAL
4 RestrictedKrbHost/ubuntu.tailabbabc.ts.net@KACIP.LOCAL
4 RestrictedKrbHost/ubuntu.tailabbabc.ts.net@KACIP.LOCAL
4 RestrictedKrbHost/ubuntu.tailabbabc.ts.net@KACIP.LOCAL
ecah@ubuntu:~$ sudo systemctl restart sssd
ecah@ubuntu:~$ sudo systemctl status sssd
● sssd.service - System Security Services Daemon
   Loaded: loaded (/usr/lib/systemd/system/sss.service)
   Active: active (running) since Wed 2026-06-10 08:25:06
 Main PID: 9743 (sss)
    Tasks: 5 (limit: 4597)
  Memory: 60.4M (peak: 60.9M)
     CPU: 279ms
    CGroup: /system.slice/sss.service
            └─9743 /usr/sbin/sss -i --logger=files
              └─9747 /usr/libexec/sss/sss_be --domain ka
                └─9748 /usr/libexec/sss/sss_nss --uid 0 --
                  └─9749 /usr/libexec/sss/sss_pam --uid 0 --
                    └─9750 /usr/libexec/sss/sss_pac --uid 0 --

Jun 10 08:25:06 ubuntu systemd[1]: Starting sss.service
Jun 10 08:25:06 ubuntu sss[9743]: Starting up
Jun 10 08:25:06 ubuntu sss_be[9747]: Starting up
Jun 10 08:25:06 ubuntu sss_pac[9750]: Starting up
Jun 10 08:25:06 ubuntu sss_nss[9748]: Starting up
Jun 10 08:25:06 ubuntu sss_pam[9749]: Starting up
Jun 10 08:25:07 ubuntu systemd[1]: Started sss.service

ecah@ubuntu:~$ id john@kacip.local
uid=1191401105(john@kacip.local) gid=1191400513(domain users@kacip.local) groups=1191400513(domain users@kacip.local)
ecah@ubuntu:~$

```

Figure 3.2.1: Verification of Active Directory domain integration by authenticating the user account johnkacip.local on Ubuntu

3.3 IDS Installation and Configuration

An Intrusion Detection System (IDS) was installed on Ubuntu to monitor network activities. The IDS was configured to monitor incoming and outgoing network traffic, detect SSH brute-force attacks, detect suspicious connection attempts and generate security alerts for malicious behaviour. The IDS continuously analyzes traffic against predefined rules and signatures. Whenever suspicious activity is detected, alerts are generated and stored in IDS log files.

```
alice@alice-VMware-Virtual-Platform:~/Desktop$ sudo systemctl restart suricata
alice@alice-VMware-Virtual-Platform:~/Desktop$ sudo systemctl enable suricata
alice@alice-VMware-Virtual-Platform:~/Desktop$ wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.8.2-1_amd64.deb && sudo WAZUH_MANAGER='100.117.105.36' WAZUH_AGENT_NAME='IDS' dpkg -i ./wazuh-agent_4.8.2-1_amd64.deb
--2026-06-14 02:25:33-- https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.8.2-1_amd64.deb
Resolving packages.wazuh.com (packages.wazuh.com)... 18.161.179.8, 18.161.179.62, 18.161.179.59, ...
Connecting to packages.wazuh.com (packages.wazuh.com)|18.161.179.8|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 10260268 (9.8M) [application/vnd.debian.binary-package]
Saving to: 'wazuh-agent_4.8.2-1_amd64.deb'

wazuh-agent_4.8.2-1_amd64.d 100%[=====>] 9.78M 783KB/s in 12s

2026-06-14 02:25:46 (840 KB/s) - 'wazuh-agent_4.8.2-1_amd64.deb' saved [10260268/10260268]

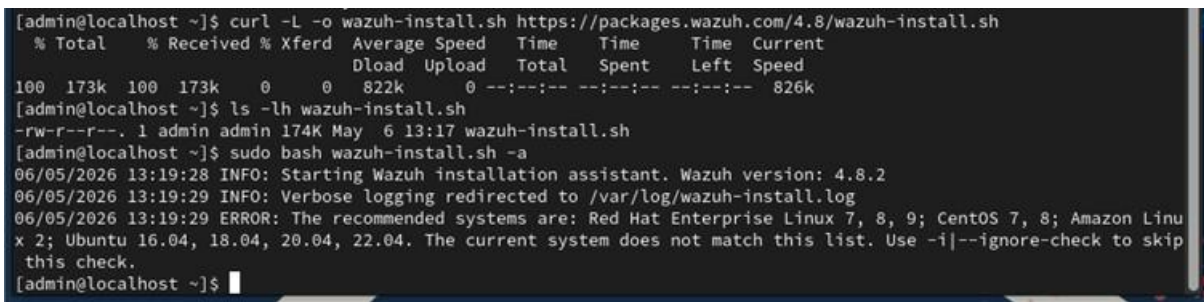
Selecting previously unselected package wazuh-agent.
(Reading database... 191989 files and directories currently installed.)
Preparing to unpack ./wazuh-agent_4.8.2-1_amd64.deb...
Unpacking wazuh-agent (4.8.2-1)...
Setting up wazuh-agent (4.8.2-1)...
alice@alice-VMware-Virtual-Platform:~/Desktop$ sudo systemctl daemon-reload
alice@alice-VMware-Virtual-Platform:~/Desktop$ sudo systemctl enable wazuh-agent
Created symlink '/etc/systemd/system/multi-user.target.wants/wazuh-agent.service' -> '/usr/lib/systemd/system/wazuh-agent.service'.
alice@alice-VMware-Virtual-Platform:~/Desktop$ sudo systemctl start wazuh-agent
alice@alice-VMware-Virtual-Platform:~/Desktop$ sudo nano /var/ossec/etc/ossec.conf
alice@alice-VMware-Virtual-Platform:~/Desktop$ sudo systemctl restart wazuh-agent
alice@alice-VMware-Virtual-Platform:~/Desktop$
```

3.4 Wazuh Deployments

The Wazuh Manager was deployed to provide centralized security monitoring. The Ubuntu endpoint was registered as a Wazuh Agent and configured to monitor system logs, authentication logs, SSH logs, IDS alerts and Sudo activity logs. The Wazuh Agents forwards collected events to the Wazuh Manager for analysis and visualization.

Because Rocky Linux is an enterprise-grade RHEL derivative, the automated installation script requires a forced architecture check bypass.

Deployment Execution Commands:



```
[admin@localhost ~]$ curl -L -o wazuh-install.sh https://packages.wazuh.com/4.8/wazuh-install.sh
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
100 173k 100 173k    0     0  822k      0 --:--:-- --:--:-- --:--:-- 826k
[admin@localhost ~]$ ls -lh wazuh-install.sh
-rw-r--r--. 1 admin admin 174K May  6 13:17 wazuh-install.sh
[admin@localhost ~]$ sudo bash wazuh-install.sh -a
06/05/2026 13:19:28 INFO: Starting Wazuh installation assistant. Wazuh version: 4.8.2
06/05/2026 13:19:29 INFO: Verbose logging redirected to /var/log/wazuh-install.log
06/05/2026 13:19:29 ERROR: The recommended systems are: Red Hat Enterprise Linux 7, 8, 9; CentOS 7, 8; Amazon Linu
x 2; Ubuntu 16.04, 18.04, 20.04, 22.04. The current system does not match this list. Use -i|--ignore-check to skip
this check.
[admin@localhost ~]$
```

1. Fetching the official Wazuh installation package and node manifest

```
curl -L -o wazuh-install.sh https://packages.wazuh.com/4.8/wazuh-install.sh
```

```
curl -A "Mozilla/5.0" -L -o config.yml https://packages.wazuh.com/4.8/config.yml
```

```
[admin@localhost ~]$ sudo bash wazuh-install.sh -a -i
06/05/2026 13:21:19 INFO: Starting Wazuh installation assistant. Wazuh version: 4.8.2
06/05/2026 13:21:19 INFO: Verbose logging redirected to /var/log/wazuh-install.log
06/05/2026 13:21:23 WARNING: Hardware and system checks ignored.
06/05/2026 13:21:31 INFO: Wazuh web interface port will be 443.
06/05/2026 13:21:33 WARNING: The system has Firewalld enabled. Please ensure that traffic is allowed on these ports: 1515, 1514, 443.
06/05/2026 13:21:35 INFO: Wazuh repository added.
06/05/2026 13:21:35 INFO: --- Configuration files ---
06/05/2026 13:21:35 INFO: Generating configuration files.
06/05/2026 13:21:36 INFO: Generating the root certificate.
06/05/2026 13:21:36 INFO: Generating Admin certificates.
06/05/2026 13:21:36 INFO: Generating Wazuh indexer certificates.
06/05/2026 13:21:36 INFO: Generating Filebeat certificates.
06/05/2026 13:21:36 INFO: Generating Wazuh dashboard certificates.
06/05/2026 13:21:37 INFO: Created wazuh-install-files.tar. It contains the Wazuh cluster key, certificates, and passwords necessary for installation.
06/05/2026 13:21:37 INFO: --- Wazuh indexer ---
06/05/2026 13:21:37 INFO: Starting Wazuh indexer installation.
06/05/2026 13:24:14 INFO: Wazuh indexer installation finished.
06/05/2026 13:24:14 INFO: Wazuh indexer post-install configuration finished.
06/05/2026 13:24:14 INFO: Starting service wazuh-indexer.
06/05/2026 13:24:38 INFO: wazuh-indexer service started.
06/05/2026 13:24:38 INFO: Initializing Wazuh indexer cluster security settings.
06/05/2026 13:24:49 INFO: Wazuh indexer cluster security configuration initialized.
06/05/2026 13:24:49 INFO: Wazuh indexer cluster initialized.
06/05/2026 13:24:49 INFO: --- Wazuh server ---
06/05/2026 13:24:49 INFO: Starting the Wazuh manager installation.
06/05/2026 13:26:22 INFO: Wazuh manager installation finished.
06/05/2026 13:26:22 INFO: Wazuh manager vulnerability detection configuration finished.
06/05/2026 13:26:22 INFO: Starting service wazuh-manager.
06/05/2026 13:26:36 INFO: wazuh-manager service started.
06/05/2026 13:26:36 INFO: Starting Filebeat installation.
06/05/2026 13:26:57 INFO: Filebeat installation finished.
```

2. Executing deployment with the compatibility override flag

sudo bash wazuh-install.sh -a -i

Configuration Details (config.yml Analysis): To establish this VM as a centralized manager, the loopback addresses (127.0.0.1) in the deployment manifest were replaced with the static Tailscale IP (100.84.226.76). This forces the Wazuh Indexer, Server, and Dashboard APIs to bind to the VPN interface, allowing cross-network log ingestion.

4. Testing scenarios and results

ROCKY LINUX

Section 1: Centralized Security Watchtower Infrastructure (Rocky Linux)

4.1 Virtual Machine & Network Topology

- **Operating System:** Rocky Linux 9 (Minimal Install)
- **Hypervisor:** VMware Workstation / Player
- **Network Adapter Configuration:** NAT mode (with Tailscale overlay network).
- **Tailscale IP Assigned:** 100.84.226.76

```
[admin@localhost ~]$ tailscale status
100.84.226.76  localhost-0      use4proj2@  linux    -
100.75.158.18  192             use4proj2@  linux    offline, last seen 31d ago
100.120.3.113  kali            use4proj2@  linux    offline, last seen 2d ago
100.81.65.121  ubuntu-1        use4proj2@  linux    offline, last seen 17d ago
100.93.69.88   ubuntu          use4proj2@  linux    offline, last seen 30d ago
100.77.254.121 win-lg667t5fo9v use4proj2@  windows  offline, last seen 24d ago
100.99.191.99  win-u9229pn5u74 use4proj2@  windows  active; relay "sin", tx 1322468 rx 7406908
```

Figure 4.1.0: The output of the `tailscale ip -4` or `tailscale status` command proves your Watchtower is actively connected to the team's mesh network. This verifies that the host is actively connected to the Tailscale overlay mesh network and can see all peer endpoints assigned to the **use4proj2@** domain.

By using Tailscale, the Watchtower infrastructure establishes a secure, encrypted WireGuard tunnel over the standard VMware NAT adapter configuration. This allows the Rocky Linux system to securely communicate with external endpoints across different physical networks without requiring complex port-forwarding rules on the host hypervisor.

The output confirms that the Rocky Linux Watchtower (100.84.226.76) has registered a complete routing table containing both Linux and Windows endpoints, paving the way for centralized security monitoring.

```
debug1: Next authentication method: password
john@kacip.local@100.99.191.99's password:
debug1: Authentications that can continue: publickey,password,keyboard-interacti
ve
Permission denied, please try again.
john@kacip.local@100.99.191.99's password:
debug1: Authentications that can continue: publickey,password,keyboard-interacti
ve
Permission denied, please try again.
john@kacip.local@100.99.191.99's password:
debug1: Authentications that can continue: publickey,password,keyboard-interacti
ve
debug1: No more authentication methods to try.
john@kacip.local@100.99.191.99: Permission denied (publickey,password,keyboard-i
nteractive).
ecah@ubuntu:~$
```

Figure 4.1.1: Failed Authentication Attempts during login

Jun 14, 2026		Defense Evasion,	Logon failure - Unknown user or bad		
@	T1078 T1531	Persistence, Privilege	password.	5	60122
> 16:58:11.00		Escalation, Initial Access,			
8		Impact			
data.win.system.message "An account failed to log on.					
Subject:					
Security ID: S-1-5-18					
Account Name: WIN-U9229PN5U74\$					
Account Domain: KACIP					
Logon ID: 0x3E7					
Logon Type: 8					
Account For Which Logon Failed:					
Security ID: S-1-0-0					
Account Name: john@kacip.local					
Account Domain:					
Failure Information:					
Failure Reason: Unknown user name or bad password.					
Status: 0xC000006D					
Sub Status: 0xC000006A					
Process Information:					
Caller Process ID: 0x13d0					
Caller Process Name: C:\Windows\System32\OpenSSH\sshd.exe					
Network Information:					
Workstation Name: WIN-U9229PN5U74					
Source Network Address: -					
Source Port: -					

Figure 4.1.2: This figure shows a high-fidelity Wazuh SIEM log capturing an authentication failure (Event ID 4625) on the Ubuntu sites where a remote system attempted to log into the Active Directory account john@kacip.local over an SSH connection using an incorrect password.

```
john@kacip.local@ubuntu:~$ ls -la /dev/shm /var/tmp /tmp
/dev/shm:
total 0
drwxrwxrwt 2 root root 40 Jun 10 07:10 .
drwxr-xr-x 19 root root 4200 Jun 10 07:10 ..

/tmp:
total 96
drwxrwxrwt 18 root root 4096 Jun 10 09:17 .
drwxr-xr-x 23 root root 4096 May 7 08:27 ..
drwxrwxrwt 2 root root 4096 Jun 10 07:10 .font-unix
drwxrwxrwt 2 root root 4096 Jun 10 07:10 .ICE-unix
-rw-r--r-- 1 ecac ecac 1526 Jun 10 08:05 krb5cc_1000
-rw-r--r-- 1 john@kacip.local domain users@kacip.local 1345 Jun 10 09:17 krb5cc_1191401105_5Ry5lR
```

Figure 4.1.3: Successful access to the Ubuntu system

The screenshot shows the Wazuh SIEM dashboard interface. At the top, there's a navigation bar with a 'Threat Hunting' tab selected. Below this, a table titled 'Security Alerts' displays a single alert. The table has columns for Time, Agent, Agent name, Technique(s), Tactics(s), Description, Level, and Rule ID. The alert is dated 'Jun 14, 2026 @ 16:55:03.740', from agent '001', and identifies the endpoint as 'WIN-U9229PN5U74'. It lists technique 'T1078' and tactics 'Defense Evasion, Persistence, Privilege Escalation, Initial Access'. The description notes 'Windows logon success'. The alert is categorized as Level 3 with rule ID '60106'.

Time ↓	Agent	Agent name	Technique(s)	Tactics(s)	Description	Level	Rule ID
Jun 14, 2026 @ 16:55:03.740	001	WIN-U9229PN5U74	T1078	Defense Evasion, Persistence, Privilege Escalation, Initial Access	Windows logon success.	3	60106

Figure 4.1.4: This figure shows the Wazuh SIEM dashboard detecting a successful Windows login event on endpoint WIN-U9229PN5U74, which is flagged under Level 3 severity to help security analysts track valid account usage for potential defense evasion or persistence tactics.

4.2 Brute-Force Attack

```
john@kacip.local@ubuntu:~$ ls -la /dev/shm /var/tmp /tmp
/dev/shm:
total 0
drwxrwxrwt  2 root root   40 Jun 10 07:10 .
drwxr-xr-x 19 root root 4200 Jun 10 07:10 ..

/tmp:
total 96
drwxrwxrwt 18 root          root          4096 Jun 10 09:17 .
drwxr-xr-x 23 root          root          4096 May  7 08:27 ..
drwxrwxrwt  2 root          root          4096 Jun 10 07:10 .font-unix
drwxrwxrwt  2 root          root          4096 Jun 10 07:10 .ICE-unix
-rw-----  1 ecah          ecah          1526 Jun 10 08:05 krb5cc_1000
-rw-----  1 john@kacip.local domain users@kacip.local 1345 Jun 10 09:17 krb5cc_1191401105_5Ry5lR
```

Figure 4.2.0: Successful access to the Ubuntu system using the domain account johnkacip.local after a brute-force attack simulation

```
ecah@ubuntu:~/Desktop/attack$ crackmapexec smb 100.99.191.99 -u john -p passwords.txt
[*] First time use detected
[*] Creating home directory structure
[*] Creating missing folder logs
[*] Creating missing folder modules
[*] Creating missing folder protocols
[*] Creating missing folder workspaces
[*] Creating missing folder obfuscated_scripts
[*] Creating missing folder screenshots
[*] Copying default configuration file
SMB 100.99.191.99 445 WIN-U9229PNSU74 [*] Windows 10.0 Build 17763 x64 (name:WIN-U9229PNSU74) (domain:kacip.local) (signing:True) (SMBv1:False)
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@1 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@2 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@3 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@4 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@5 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@6 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@7 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@8 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@9 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@10 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@11 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john: STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@13 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@14 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@15 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@16 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@17 STATUS_LOGON_FAILURE
```

Figure 4.2.1: Execution of a brute-force attack using CrackMapExec against the Active Directory domain account john through the SMB service

```
p.local) (signing:True) (SMBv1:False)
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@1 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@2 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@3 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@4 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@5 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@6 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@7 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@8 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@9 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@10 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@11 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john: STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@13 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@14 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@15 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@16 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@17 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@18 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@19 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@20 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@21 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@22 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@23 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@24 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@25 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@26 STATUS_LOGON_FAILURE
SMB 100.99.191.99 445 WIN-U9229PNSU74 [-] kacip.local\john:Abc@27 STATUS_LOGON_FAILURE
^Cecah@ubuntu:~/Desktop/attack
```

Figure 4.2.2: Repeated failed SMB login attempts against the Active Directory account johnkacip.local during a brute-force attack simulation

```
^Cecah@ubuntu:~/Desktop/attackhydra -l john@kacip.local -P passwords.txt ssh://100.99.191.99
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations,
or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-06-14 08:54:58
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 51 login tries (l:1/p:51), ~4 tries per task
[DATA] attacking ssh://100.99.191.99:22/
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-06-14 08:55:34
```

Figure 4.2.3: Hydra successfully identified a valid SSH credential for the domain account johnkacip.local during the brute-force attack simulation

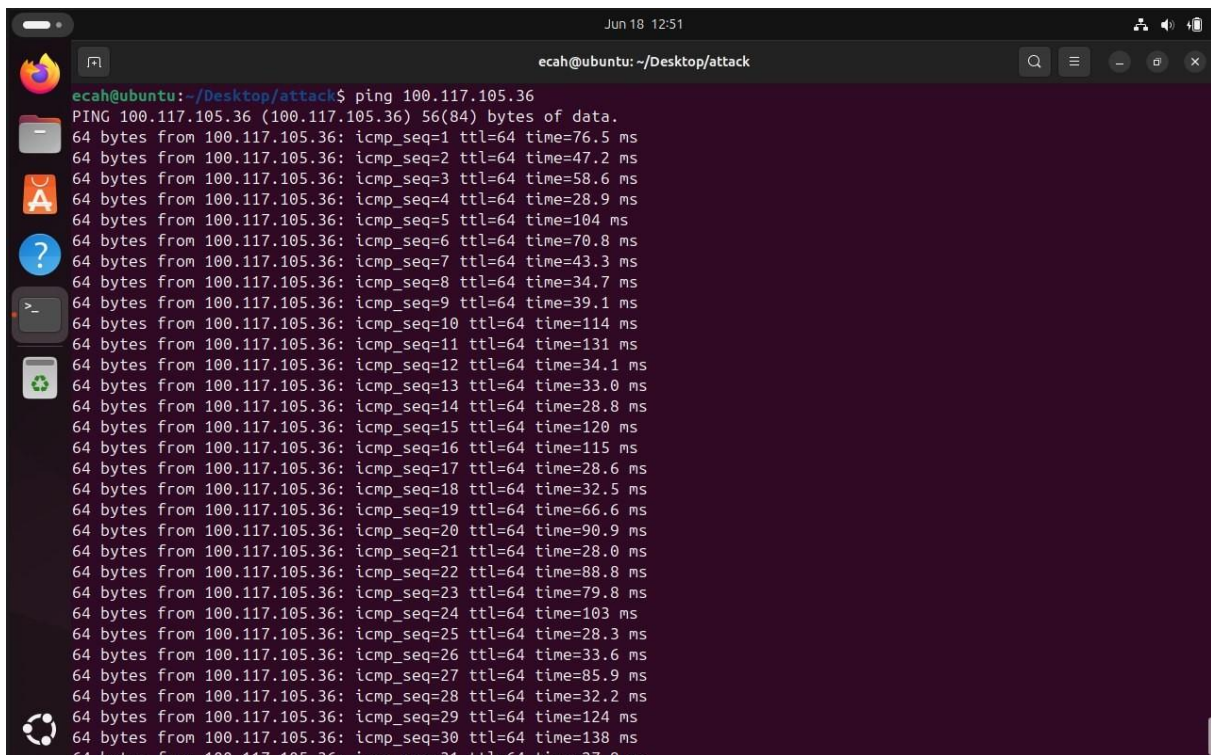
Jun 18, 2026							
12:50:57.006	001	WIN-U9229PN5U74	T1110	Credential Access	Multiple Windows logon failures.	10	60204
rule.description	Multiple Windows logon failures.						
rule.firedtimes	6						
rule.frequency	8						
rule.gdpr	IV_32.2, IV_35.7.d						
rule.groups	windows, windows_security, authentication_failures						
rule.hipaa	164.312.b						
rule.id	60204						
rule.level	10						
rule.mail	false						
rule.mitre.id	T1110						
rule.mitre.tactic	Credential Access						
rule.mitre.technique	Brute Force						
rule.nist_800_53	AC.7, AU.14, SI.4						
rule.pci_dss	10.2.4, 10.2.5, 11.4						
rule.tsc	CC6.1, CC6.8, CC7.2, CC7.3						
timestamp	2026-06-18T12:50:57.006+0800						

Figure 4.2.4: A high-severity security alert captured by Wazuh SIEM dashboard, monitoring the active Windows endpoint we saw earlier in your Tailscale status (win-u9229pn5u74).

The SIEM triggered **Rule ID 60204**, which indicates **"Multiple Windows logon failures."** This means someone (or an automated script) tried to log into the Windows machine multiple times with wrong credentials. The victim host is WIN-U9229PN5U74 (the active Windows node connected via Tailscale).

The alert fired because the Wazuh agent recorded a rapid succession of failed logon events (rule.firedtimes: 6). This type of telemetry points directly to an automated password-spraying or brute-force tool attempting to crack local user accounts on the Windows machine.

4.3 IDS ATTACKED

A terminal window titled 'ecah@ubuntu: ~/Desktop/attack' with a date and time of 'Jun 18 12:51'. The terminal shows a continuous ping command being executed: 'ecah@ubuntu:~/Desktop/attack\$ ping 100.117.105.36'. The output consists of 31 lines of ping results, each showing '64 bytes from 100.117.105.36: icmp_seq=X ttl=64 time=Y ms' where X ranges from 1 to 31 and Y represents the round-trip time in milliseconds. The times vary between approximately 27.0 ms and 138 ms. The terminal window has a dark background and a sidebar on the left with various application icons.

```
Jun 18 12:51
ecah@ubuntu: ~/Desktop/attack
ecah@ubuntu:~/Desktop/attack$ ping 100.117.105.36
PING 100.117.105.36 (100.117.105.36) 56(84) bytes of data.
64 bytes from 100.117.105.36: icmp_seq=1 ttl=64 time=76.5 ms
64 bytes from 100.117.105.36: icmp_seq=2 ttl=64 time=47.2 ms
64 bytes from 100.117.105.36: icmp_seq=3 ttl=64 time=58.6 ms
64 bytes from 100.117.105.36: icmp_seq=4 ttl=64 time=28.9 ms
64 bytes from 100.117.105.36: icmp_seq=5 ttl=64 time=104 ms
64 bytes from 100.117.105.36: icmp_seq=6 ttl=64 time=70.8 ms
64 bytes from 100.117.105.36: icmp_seq=7 ttl=64 time=43.3 ms
64 bytes from 100.117.105.36: icmp_seq=8 ttl=64 time=34.7 ms
64 bytes from 100.117.105.36: icmp_seq=9 ttl=64 time=39.1 ms
64 bytes from 100.117.105.36: icmp_seq=10 ttl=64 time=114 ms
64 bytes from 100.117.105.36: icmp_seq=11 ttl=64 time=131 ms
64 bytes from 100.117.105.36: icmp_seq=12 ttl=64 time=34.1 ms
64 bytes from 100.117.105.36: icmp_seq=13 ttl=64 time=33.0 ms
64 bytes from 100.117.105.36: icmp_seq=14 ttl=64 time=28.8 ms
64 bytes from 100.117.105.36: icmp_seq=15 ttl=64 time=120 ms
64 bytes from 100.117.105.36: icmp_seq=16 ttl=64 time=115 ms
64 bytes from 100.117.105.36: icmp_seq=17 ttl=64 time=28.6 ms
64 bytes from 100.117.105.36: icmp_seq=18 ttl=64 time=32.5 ms
64 bytes from 100.117.105.36: icmp_seq=19 ttl=64 time=66.6 ms
64 bytes from 100.117.105.36: icmp_seq=20 ttl=64 time=90.9 ms
64 bytes from 100.117.105.36: icmp_seq=21 ttl=64 time=28.0 ms
64 bytes from 100.117.105.36: icmp_seq=22 ttl=64 time=88.8 ms
64 bytes from 100.117.105.36: icmp_seq=23 ttl=64 time=79.8 ms
64 bytes from 100.117.105.36: icmp_seq=24 ttl=64 time=103 ms
64 bytes from 100.117.105.36: icmp_seq=25 ttl=64 time=28.3 ms
64 bytes from 100.117.105.36: icmp_seq=26 ttl=64 time=33.6 ms
64 bytes from 100.117.105.36: icmp_seq=27 ttl=64 time=85.9 ms
64 bytes from 100.117.105.36: icmp_seq=28 ttl=64 time=32.2 ms
64 bytes from 100.117.105.36: icmp_seq=29 ttl=64 time=124 ms
64 bytes from 100.117.105.36: icmp_seq=30 ttl=64 time=138 ms
64 bytes from 100.117.105.36: icmp_seq=31 ttl=64 time=27.0 ms
```

Figure 4.3.0: This figure illustrates the reconnaissance phase where the Ubuntu endpoint (ecah@ubuntu) initiates continuous ICMP echo requests (ping) against the target IP address 100.117.105.36 to verify host availability before launching active IDS simulation testing.

86601

data.alert.gid	1
data.alert.rev	2
data.alert.severity	2
data.alert.signature	LOCAL PING DETECTED
data.alert.signature_id	9999991
data.dest_ip	100.99.191.99
data.direction	to_server
data.event_type	alert
data.flow.bytes_to_client	0
data.flow.bytes_to_server	84
data.flow.dest_ip	100.99.191.99
data.flow.pkts_to_client	0
data.flow.pkts_to_server	1
data.flow.src_ip	100.117.105.36

Figure 4.3.1: The security alert log generated by an Intrusion Detection System (IDS) named Suricata. It proves that a system on the network sent a network ping (ICMP request) to a server, and a custom security rule was triggered to flag that activity.

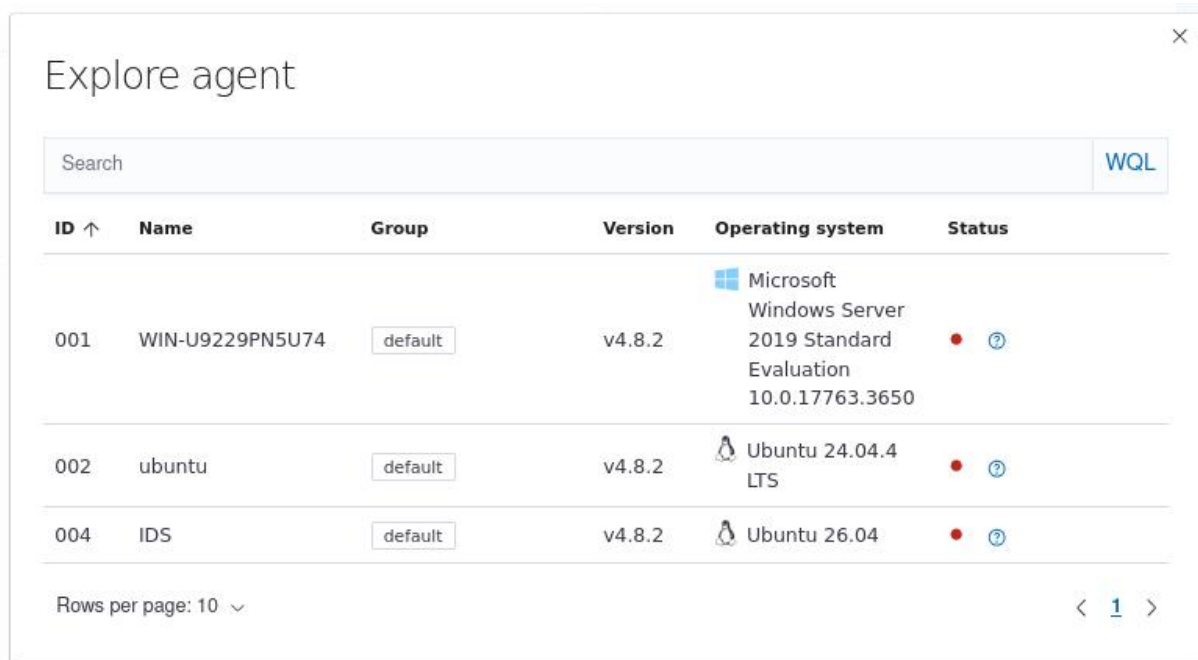
Section 2: Log Ingestion & Endpoint Verification

2.1 Centralized Dashboard Access

The security console is reached securely across the Tailscale network via HTTPS: <https://100.84.226.76>.

2.2 Active Agent Status Verification

To verify that the team's infrastructure is reporting correctly to the central SIEM, navigate to: **Wazuh Dashboard > Menu (≡) > Management > Administration > Agents**



The screenshot shows the 'Explore agent' window in the Wazuh Dashboard. It features a search bar at the top with a 'WQL' button. Below the search bar is a table with columns: ID ↑, Name, Group, Version, Operating system, and Status. The table lists three agents: 001 (WIN-U9229PN5U74), 002 (ubuntu), and 004 (IDS). Each agent has a 'default' group, version 'v4.8.2', and a status of 'Online' (indicated by a red dot and a question mark icon). The operating system details are: Microsoft Windows Server 2019 Standard Evaluation (10.0.17763.3650) for agent 001, Ubuntu 24.04.4 LTS for agent 002, and Ubuntu 26.04 for agent 004. At the bottom, there is a 'Rows per page: 10' dropdown and pagination controls showing page 1 of 1.

ID ↑	Name	Group	Version	Operating system	Status
001	WIN-U9229PN5U74	default	v4.8.2	Microsoft Windows Server 2019 Standard Evaluation 10.0.17763.3650	Online
002	ubuntu	default	v4.8.2	Ubuntu 24.04.4 LTS	Online
004	IDS	default	v4.8.2	Ubuntu 26.04	Online

Figure 4.2.0 shown Agent Table

1. The **Windows Server Active Directory** endpoint (Ingesting Windows Security Event Logs).
2. The **Ubuntu Endpoint** (Ingesting Linux Authentication Logs).
3. The **IDS** endpoint is (Ingesting Network Traffic Logs / Network Security Alerts).

5. Security analysis and discussion

The implemented solution demonstrates a layered security architecture that combines centralized authentication, intrusion detection, and SIEM-based monitoring.

Active Directory provides centralized identity management and reduces administrative complexity by maintaining a single authentication source. Ubuntu domain integration eliminates the need for separate local user accounts while ensuring consistent access control policies.

The IDS serves as the first line of defense by actively monitoring network traffic and identifying suspicious activities. Unlike standard logging mechanisms, the IDS performs traffic inspection and pattern analysis to detect attacks in real time.

Wazuh enhances security visibility by collecting logs from multiple sources and correlating related events. This allows administrators to quickly identify attack patterns, investigate incidents, and respond to threats efficiently.

The testing scenarios confirmed that both network-based attacks and privilege escalation attempts could be detected and analyzed effectively.

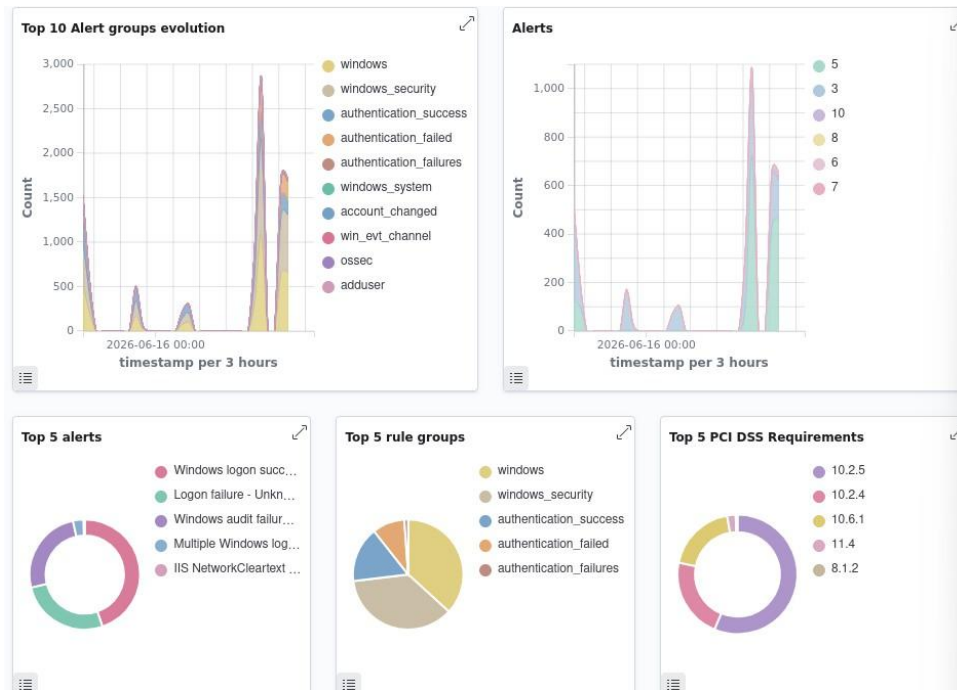


Figure 5.1.0: This figure displays the overall Wazuh SIEM analytics dashboard, capturing significant traffic spikes that correspond to automated authentication attacks, alongside structural breakdowns of rule severities, top security alerts, and associated PCI DSS regulatory compliance mappings.

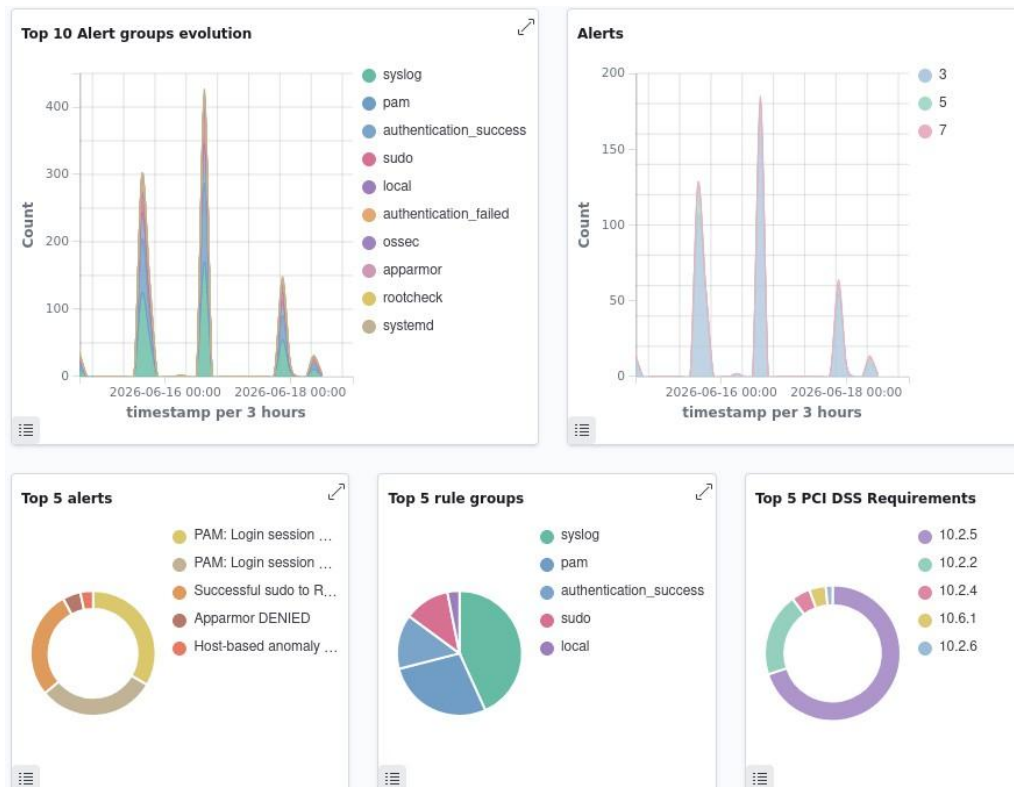


Figure 5.1.2: This figure shows the Wazuh SIEM overview for the Ubuntu agent providing visual proof of successful monitoring for Linux authentication states, AppArmor policy enforcements, and successful root privilege escalations (sudo), automatically mapped to PCI DSS regulatory compliance criteria.

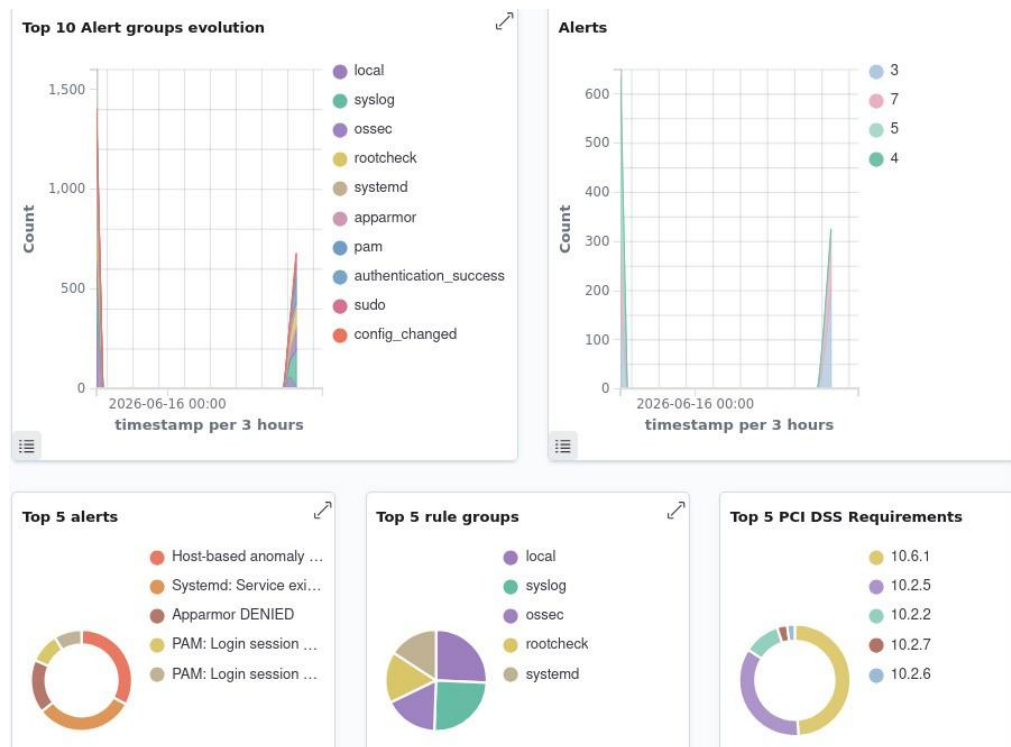


Figure 5.1.3: This figure displays the Wazuh analytics overview for the dedicated IDS agent proving the system's ability to aggregate host-based anomaly data, trace rule-tuning events (config_changed), and map network telemetry directly to PCI DSS log review standard 10.6.1.

6. Challenges and Limitations

6.1 Challenges

```
[admin@localhost ~]$ sudo bash wazuh-install.sh -a -i
06/05/2026 13:21:19 INFO: Starting Wazuh installation assistant. Wazuh version: 4.8.2
06/05/2026 13:21:19 INFO: Verbose logging redirected to /var/log/wazuh-install.log
06/05/2026 13:21:23 WARNING: Hardware and system checks ignored.
06/05/2026 13:21:31 INFO: Wazuh web interface port will be 443.
06/05/2026 13:21:33 WARNING: The system has Firewalld enabled. Please ensure that traffic is allowed on these ports: 1515, 1514, 443.
06/05/2026 13:21:35 INFO: Wazuh repository added.
06/05/2026 13:21:35 INFO: --- Configuration files ---
06/05/2026 13:21:35 INFO: Generating configuration files.
06/05/2026 13:21:36 INFO: Generating the root certificate.
06/05/2026 13:21:36 INFO: Generating Admin certificates.
06/05/2026 13:21:36 INFO: Generating Wazuh indexer certificates.
06/05/2026 13:21:36 INFO: Generating Filebeat certificates.
06/05/2026 13:21:36 INFO: Generating Wazuh dashboard certificates.
06/05/2026 13:21:37 INFO: Created wazuh-install-files.tar. It contains the Wazuh cluster key, certificates, and passwords necessary for installation.
06/05/2026 13:21:37 INFO: --- Wazuh indexer ---
06/05/2026 13:21:37 INFO: Starting Wazuh indexer installation.
06/05/2026 13:24:14 INFO: Wazuh indexer installation finished.
06/05/2026 13:24:14 INFO: Wazuh indexer post-install configuration finished.
06/05/2026 13:24:14 INFO: Starting service wazuh-indexer.
06/05/2026 13:24:38 INFO: wazuh-indexer service started.
06/05/2026 13:24:38 INFO: Initializing Wazuh indexer cluster security settings.
06/05/2026 13:24:49 INFO: Wazuh indexer cluster security configuration initialized.
06/05/2026 13:24:49 INFO: Wazuh indexer cluster initialized.
06/05/2026 13:24:49 INFO: --- Wazuh server ---
06/05/2026 13:24:49 INFO: Starting the Wazuh manager installation.
06/05/2026 13:26:22 INFO: Wazuh manager installation finished.
06/05/2026 13:26:22 INFO: Wazuh manager vulnerability detection configuration finished.
06/05/2026 13:26:22 INFO: Starting service wazuh-manager.
06/05/2026 13:26:36 INFO: wazuh-manager service started.
06/05/2026 13:26:36 INFO: Starting Filebeat installation.
06/05/2026 13:26:57 INFO: Filebeat installation finished.
```

Several key technical challenges were encountered and successfully resolved during the implementation phase of the security infrastructure:

- The primary challenge involved integrating Linux authentication with Active Directory. User accounts initially failed to map correctly because POSIX attributes were not configured. This issue was resolved by manually assigning uidNumber and gidNumber values.
- IDS tuning also required adjustments to minimize false positives and ensure relevant security alerts were generated. The detection rules produced an overwhelming volume of alerts for benign network activities, meaning the rulesets had to be methodically tailored to the laboratory environment to isolate genuine threats like brute-force attempts from standard administrative background noise.
- A significant infrastructure hurdle arose during the initial verification phase of the SIEM deployment script. As captured at timestamp 13:21:23 in figure above, the installation assistant flagged a warning because hardware and

system pre-checks had to be explicitly ignored using the `-i` flag. Because the virtual laboratory environment fell below the strict production resource thresholds required by the Wazuh stack, the installation had to be forced, which subsequently demanded careful monitoring to ensure the Java-based indexer services did not exhaust virtual memory and crash the Rocky Linux host.

- Managing the automated generation of TLS certificates presented an added layer of complexity. As shown between timestamps 13:21:35 and 13:21:37 in figure above, the script had to mint unique certificates for the root CA, admin, indexer, Filebeat, and dashboard. These had to be perfectly bundled into the `wazuh-install-files.tar` archive; a single cryptographic mismatch during the cluster initialization at 13:24:38 would break all internal SIEM communications.

6.2 Limitations

Several limitations were identified during the project:

- The IDS occasionally generated alerts for legitimate network activities. Although rule tuning was performed, false positives may still occur and could increase the workload of security administrators.
- Testing was mainly focused on SSH and SMB brute-force attacks. Other attack types, such as malware, ransomware, and web application attacks, were not evaluated during the project.
- While Wazuh successfully detected security events, response actions were performed manually. This may result in slower reaction times when handling security incidents.

7. Conclusion and Recommendations

7.1 Conclusion

This project successfully implemented a centralized security monitoring environment by integrating Active Directory, Ubuntu Linux, an Intrusion Detection System (IDS), and the Wazuh Security Information and Event Management (SIEM) platform. The integration of these technologies provided centralized authentication, real-time network monitoring, and centralized log analysis within a multi-platform environment.

The Ubuntu system was successfully joined to the Active Directory domain, allowing users to authenticate using a single set of credentials. In addition, the IDS effectively monitored network traffic and generated alerts when suspicious activities were detected. These alerts, together with authentication and system logs, were collected and analyzed by Wazuh, providing administrators with improved visibility into security events.

The testing results demonstrated that the implemented solution was capable of detecting authentication failures, brute-force attacks, and suspicious network activities. The combination of Active Directory, IDS, and Wazuh created a layered security approach that improved threat detection and monitoring capabilities. The project achieved its objectives and demonstrated the importance of centralized authentication and security monitoring in enhancing an organization's cybersecurity posture.

7.2 Recommendations

To further improve the security environments, the following recommendations are purposed:

- Deploy additional Linux and Windows endpoints.
- Implement Multi-Factor Authentication (MFA).
- Integrate endpoint protection and antivirus solutions.
- Configure automated incident response actions.
- Enable threat intelligence integration within Wazuh.
- Conduct periodic penetration testing and security audits.
- Expands IDS rule sets to improve attack detection coverage.

The implementation demonstrates a practical enterprise security architecture capable of providing centralized authentication, intrusion detection, and security event monitoring within a multi-platform environment.

Link YouTube:

<https://youtu.be/BklrJjUY2ac?si=7sEbSp-e0r93e9DX>